



Work Done By Spring

$$W_{\text{top}} = \int_{\Delta x} F_s dx$$

$$W_{\text{bottom}} = - \int_{\Delta x} F_s dx$$

the force is now pointed in the negative direction. so negative displacement and positive force must produce positive work.

$$W = \int_{\Delta x} F_s dx_{\text{top}} - \int_{\Delta x} F_s dx_{\text{bottom}}$$

$$W = \int_{\Delta t} (F_s V_{\text{top}} - F_s V_{\text{bottom}}) dt$$

$$W = \int_{\Delta t} F_s (V_{\text{top}} - V_{\text{bottom}}) dt$$

for approximation:

$$W = \sum F_s(t) (dX_{\text{top}}(t) - dX_{\text{bottom}}(t))$$

$F_s(t)$:

$$F_s - F_g = m a_{\text{top}}(t) \Rightarrow F_s = F_g + m a_{\text{top}}(t)$$

$$F_s(t) = m(g + a_{\text{top}}(t))$$

$dX(t)$:

$$V(t_n) = V(t_{n-1}) + \frac{1}{2} (a_{n-1} + a_n) (t_n - t_{n-1})$$

use to make array $V(t)$

$$dX(t_n) = \frac{1}{2} (V(t_{n-1}) + V(t_n)) (t_n - t_{n-1})$$

use to make array $dX(t)$